

Least-squares regression

Multiple, multivariate, PCR and PLS approaches

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Linear regression

- ▶ Simple regression: one predictor, one response
 - ▶ Typical notation: $Y = \beta_0 + \beta_1 X + \epsilon$
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 - ▶ Example: House price against the number of rooms, the size of the garden, the distance to the city center, etc.

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 - ▶ Example: House price against the number of rooms, the size of the garden, the distance to the city center, etc.
- ▶ Multivariate regression: multiple predictors, multiple responses
 - ▶ Typical notation: $\mathcal{Y} = \mathbf{X}'\mathcal{B} + \mathcal{E}$, where $\mathcal{Y}$, $\mathcal{B}$ and $\mathcal{E}$ have r columns
 - ▶ Example: Rainfall and wind speed against hurricane category, altitude and distance from the coast

Transformations of input variables

- ▶ We've often transformed input variables individually, e.g. squaring them or taking the logarithm
 - ▶ For example, if we have a predictor X , we can use X^2 as a predictor and have
$$Y = \beta_0 + \beta_1 X^2 + \epsilon$$

Transformations of input variables

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 - ▶ For example, if we have a predictor X , we can use X^2 as a predictor and have
$$Y = \beta_0 + \beta_1 X^2 + \epsilon$$
- ▶ We will also now consider taking *linear combinations* of different input variables, e.g. creating the variable $X_1^* = \alpha_1 X_1 + \alpha_2 X_2$ and using it as a predictor in the model
 - ▶ We consider two ways of deciding on the linear combinations:
 - ▶ Principal components analysis: maximising variance
 - ▶ Partial least squares: maximising association with the response

A noteworthy note on notable notation

- ▶ Number of...
 - ▶ Observations: n
 - ▶ Responses: r , where $r = 1$ for multiple regression ($\mathbf{y} : n \times 1$) but $r \geq 1$ for multivariate regression ($\mathbf{Y} : n \times r$)
 - ▶ Predictors: k , i.e. $\boldsymbol{\beta} : k \times 1$ and $\mathcal{B} : k \times r$
 - ▶ Predictors, apart from the intercept: $p = k - 1$, i.e. $\mathbf{x}'\boldsymbol{\beta} = \beta_0 + \beta_1x_1 + \dots + \beta_px_p$

Observational unit

Multiple regression

- ▶ Each observational unit consists of a single response variable (y_i) and multiple predictors

$$(\mathbf{x}_i = [x_{i1} \ x_{i2} \ \dots \ x_{ip}]')$$

- ▶ Example responses per observational unit:
 - ▶ Annual revenue of a company
 - ▶ Number of pigeons at a single university
 - ▶ Blood pressure of an individual

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Multivariate regression

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$$(\mathbf{y}_i = [y_{i1} \ y_{i2} \ \dots \ y_{ir}]')$$

and multiple predictors

$$(\mathbf{x}_i = [x_{i1} \ x_{i2} \ \dots \ x_{ip}]')$$

- ▶ Example responses per observational unit:
 - ▶ Stock prices of a single company at multiple timepoints
 - ▶ Number of pigeons and starlings at a single university
 - ▶ Measurements of 1000 genes of an individual

Model assumptions

Multiple regression

- ▶ **Linearity:**

$$E[Y] = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

- ▶ **Constant variance:** $\text{Var}[\epsilon] = \sigma^2$

- ▶ **Independence:**

$$\text{Cov}[\epsilon_i, \epsilon_j] = 0 \text{ for } i \neq j$$

- ▶ Different observations entirely independent
- ▶ For example, different individuals in a study have independent BMI measurements

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Multivariate regression

- ▶ **Linearity:** $E[y] = \mathbf{X}'\mathcal{B}$

- ▶ **Constant variance:** $\text{Var}[\mathcal{E}] = \Sigma$

- ▶ **Independence:**

$$\text{Cov}[\mathcal{E}_i, \mathcal{E}_j] = \mathbf{0} \text{ for } i \neq j$$

- ▶ Different observational units are independent, but different responses within the same observational unit may be correlated
- ▶ For example, different individuals have independent BMI measurements, but multiple measurements of BMI from the same individual are clearly correlated

Estimating the regression coefficients

Multiple regression

Choose $\hat{\beta}$ to minimise the sum of squared residuals:

$$\begin{aligned} & \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= (\mathbf{y} - \hat{\mathbf{y}})'(\mathbf{y} - \hat{\mathbf{y}}), \\ &= (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}). \end{aligned}$$

This implies that

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

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This implies that

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Multivariate regression

Choose $\hat{\mathcal{B}}$ to minimise the sum of squared residuals:

$$\begin{aligned} & \sum_{i=1}^n \sum_{j=1}^r (y_{ij} - \hat{y}_{ij})^2 \\ &= \text{tr}[(\mathcal{Y} - \mathbf{X}\hat{\mathcal{B}})'(\mathcal{Y} - \mathbf{X}\hat{\mathcal{B}})]. \end{aligned}$$

This implies that

$$\hat{\mathcal{B}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

- ▶ The fitted values are the values of the response variable predicted by the model:
 - ▶ **Multiple regression:** $\hat{y}_i = \mathbf{x}'_i \hat{\beta}$
 - ▶ **Multivariate regression:** $\hat{\mathcal{Y}}_{[i]} = \mathbf{x}'_i \hat{\mathcal{B}}$
 - ▶ By convention, we use square brackets (e.g. $\mathbf{A}_{[i]}$) to denote the i -th row of a matrix

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 - ▶ By convention, we use square brackets (e.g. $\mathbf{A}_{[i]}$) to denote the i -th row of a matrix

The hat matrix

- ▶ A special matrix appears in both cases (now predicting the response for all observations at once):
 - ▶ **Multiple regression:** $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$
 - ▶ **Multivariate regression:** $\hat{\mathbf{y}} = \mathbf{X}\hat{\mathbf{B}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$

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 - ▶ **Multivariate regression:** $\hat{\mathbf{y}} = \mathbf{X}\hat{\mathbf{B}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$
- ▶ This matrix is called the **hat matrix** and is denoted by **H**
 - ▶ The reason for the name is that it “puts a hat” on the response vector $\mathbf{y}$ or y
- ▶ Special properties:
 - ▶ It is a projection matrix, i.e. $\mathbf{H}\mathbf{y} = \text{proj}_{\mathbf{X}}(\mathbf{y})$
 - ▶ It is idempotent, i.e. $\mathbf{H}^2 = \mathbf{H}$

Estimator properties

Multiple regression

- ▶ **Unbiased:** $E[\hat{\beta}] = \beta$
- ▶ **Zero-mean residual:**
 $E[\epsilon] = E[\mathbf{X}'\beta - \mathbf{X}'\hat{\beta}] = 0$
- ▶ **Correlated estimation errors:**
 $\text{Cov}[\hat{\beta}] = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$
- ▶ **Correlated residuals:** $\text{Cov}[\hat{\epsilon}] = \sigma^2(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \sigma^2(\mathbf{I} - \mathbf{H})$

Multivariate regression

- ▶ **Unbiased:** $E[\hat{\mathcal{B}}] = \mathcal{B}$
- ▶ **Zero-mean residual:**
 $E[\mathcal{E}] = E[\mathbf{X}'\mathcal{B} - \mathbf{X}'\hat{\mathcal{B}}] = 0$
- ▶ **Correlated estimation errors:**
 $\text{Cov}[\hat{\mathcal{B}}_i, \hat{\mathcal{B}}_j] = \sigma_{ij}^2(\mathbf{X}'\mathbf{X})^{-1}$ for $i, j \in \{1, \dots, r\}$, where $\mathcal{B}_i$ is the i th column of $\mathcal{B}$
- ▶ **Correlated residuals:**
 $\text{Cov}[\hat{\mathcal{E}}_i, \hat{\mathcal{E}}_j] = \sigma_{ij}(\mathbf{I} - \mathbf{H})$ for $i, j \in \{1, \dots, r\}$
 - ▶ Now the correlation is both between different responses (columns of $\mathcal{Y}$) and between different observations (rows of $\mathcal{Y}$)

Estimating the variance

Multiple regression

- ▶ The estimator of the variance is

$$\hat{\sigma}^2 = s^2 = \frac{1}{n - k} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})' (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

- ▶ Properties:
 - ▶ **Unbiased:** $E[s^2] = \sigma^2$
 - ▶ **Independent of $\hat{\boldsymbol{\beta}}$:** $\text{Cov}[s^2, \hat{\boldsymbol{\beta}}] = \mathbf{0}$

Estimating the variance

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- ▶ The estimator of the variance is

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- ▶ Properties:
 - ▶ **Unbiased:** $E[s^2] = \sigma^2$
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Multivariate regression

- ▶ The estimator of the variance Σ is

$$\hat{\Sigma} = \mathbf{S} = \frac{1}{n-k}(\mathbf{y} - \mathbf{X}\hat{\mathcal{B}})'(\mathbf{y} - \mathbf{X}\hat{\mathcal{B}}).$$

- ▶ Properties:
 - ▶ **Unbiased:** $E[\mathbf{S}] = \Sigma$
 - ▶ **Independent of $\hat{\mathcal{B}}$:** $\mathbf{S}$ and $\hat{\mathcal{B}}$ are independent

Sums of squares: Multiple regression

The total sum of squares is

$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2,$$

and can be decomposed into

$$\text{SSR} + \text{SSE} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

where SST, SSR and SSE are the sums of squares due to the total, regression and error, respectively.

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where SST, SSR and SSE are the sums of squares due to the total, regression and error, respectively.

In matrix notation, these are given by

$$\text{SST} = \mathbf{y}'\mathbf{y} - n\bar{y}^2,$$

$$\text{SSR} = \hat{\mathbf{y}}'\hat{\mathbf{y}} - n\bar{y}^2, \text{ and}$$

$$\text{SSE} = \mathbf{y}'\mathbf{y} - \hat{\mathbf{y}}'\hat{\mathbf{y}}.$$

Sums of squares: Multivariate regression

In multivariate regression, there is no universal definition of the sums of squares, but we can consider the following:

$$\text{SST} = \sum_{i=1}^n \sum_{j=1}^r (y_{ij} - \bar{y}_j)^2,$$

which can be decomposed into

$$\text{SSR} + \text{SSE} = \sum_{i=1}^n \sum_{j=1}^r (\hat{y}_{ij} - \bar{y}_j)^2 + \sum_{i=1}^n \sum_{j=1}^r (y_{ij} - \hat{y}_{ij})^2,$$

where SST, SSR and SSE are the sums of squares due to the total, regression and error, respectively.

Assessment of the model: R^2

Multiple regression

The R^2 statistic is defined as

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

- ▶ R^2 is the proportion of the total variance explained by the model
- ▶ $0 \leq R^2 \leq 1$

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- ▶ $0 \leq R^2 \leq 1$

Multivariate regression

One can envision an appropriate multivariate extension to R^2 , but it is not universally defined.

- ▶ One possibility is to consider the proportion of the total variance explained by the model for each response variable, and average this.

Note: we won't really use the sums of squares or R^2 for multivariate regression in this module.

Multiple regression example

We model the sales price of a house against its size and assessed value.

We load the data as follows:

```
data(  
  "data_tidy_house_price",  
  package = "DataTidy23RodoHonsMult"  
)
```

size	assessed_value	selling_price
15.31	57.3	74.8
15.20	63.8	74.0
16.25	65.4	72.9
14.33	57.0	70.0

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14.33	57.0	70.0

We extract the response matrix, y :

```
y_vec <- data_tidy_house_price |>  
  dplyr::pull(selling_price)
```

We extract the data for the predictor matrix, X , and prepend a column of 1s:

```
X_mat <- cbind(  
  1,  
  data_tidy_house_price |>  
    dplyr::select(-selling_price)  
) |>  
  as.matrix()  
colnames(X_mat)[1] <- "intercept"
```

Multiple regression example (cont.)

To fit the model, we first obtain the estimated regression coefficients:

```
beta_vec <- (  
  solve(t(X_mat) %*% X_mat) %*%  
  t(X_mat)  
  ) %*%  
  y_vec  
colnames(beta_vec) <- "estimate"
```

	estimate
intercept	31.000
size	2.600
assessed_value	0.045

Multiple regression example (cont.)

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```

	estimate
intercept	31.000
size	2.600
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This matches the output from the `lm` function:

```
lm(  
  selling_price ~ .,  
  data = data_tidy_house_price  
) |>  
  coef() |> signif(2) |> knitr::kable()
```

	x
(Intercept)	31.000
size	2.600
assessed_value	0.045

Multiple regression example (cont.)

To assess the model, we calculate the R^2 statistic:

```
SST <- sum((y_vec - mean(y_vec))^2)
SSE <- (y_vec - X_mat %*% beta_vec)^2 |>
  sum()
SSR <- SST - SSE
R2 <- SSR / SST
R2 |> signif(2)
```

```
[1] 0.83
```

Multiple regression example (cont.)

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R2 |> signif(2)
```

```
[1] 0.83
```

- ▶ An R^2 of 0.83 indicates that 83% of the variance in house prices is explained by the size and assessed value of the house.
- ▶ This is a high value, suggesting that the model explains the response “well”.

Multivariate example

Consider a dataset of four wood-pulp variables and four paper properties:

```
data(  
  "data_tidy_paper",  
  package = "DataTidy23RodoHonsMult"  
)
```

pulp_1	pulp_2	pulp_3	pulp_4	paper_1	paper_2	paper_3	paper_4
-0.030	35.239	36.991	1.057	21.312	7.039	5.326	0.932
0.015	35.713	36.851	1.064	21.206	6.979	5.237	0.871
0.025	39.220	30.586	1.053	20.709	6.779	5.060	0.742
0.030	39.756	21.072	1.050	19.542	6.601	4.479	0.513

Multivariate example (cont.)

We extract the response matrix, $\mathbf{Y}$:

```
Y_mat <- data_tidy_paper |>  
  dplyr::select(-starts_with("pulp")) |>  
  as.matrix()
```

paper_1	paper_2	paper_3	paper_4
21	7.0	5.3	0.93
21	7.0	5.2	0.87
21	6.8	5.1	0.74
20	6.6	4.5	0.51

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21	7.0	5.2	0.87
21	6.8	5.1	0.74
20	6.6	4.5	0.51

We extract the data for the predictor matrix, $\mathbf{X}$, and prepend a column of 1s:

```
X_mat <- as.matrix(cbind(1,
  data_tidy_paper |>
    dplyr::select(starts_with("pulp")))
))
colnames(X_mat)[1] <- "intercept"
```

intercept	pulp_1	pulp_2	pulp_3	pulp_4
1	-0.030	35	37	1.1
1	0.015	36	37	1.1
1	0.025	39	31	1.1

Multivariate example (cont.)

We first obtain the estimated regression coefficients:

```
B_mat <- (  
  solve(t(X_mat) %*% X_mat) %*%  
  t(X_mat)) %*% Y_mat
```

	paper_1	paper_2	paper_3
intercept	-74.000	-24.0000	-46.000
pulp_1	-3.100	-1.2000	-1.500
pulp_2	0.098	0.0091	0.047
pulp_3	0.049	0.0084	0.025
pulp_4	85.000	29.0000	46.000

Multivariate example (cont.)

We first obtain the estimated regression coefficients:

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B_mat <- (  
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	paper_1	paper_2	paper_3
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pulp_2	0.098	0.0091	0.047
pulp_3	0.049	0.0084	0.025
pulp_4	85.000	29.0000	46.000

This matches the output from the `lm` function (exc. fourth column for space):

```
lm(  
  cbind(paper_1, paper_2, paper_3, paper_4)  
  data = data_tidy_paper  
)$coefficients[, 1:3] |> signif(2) |>
```

	paper_1	paper_2	paper_3
(Intercept)	-74.000	-24.0000	-46.000
pulp_1	-3.100	-1.2000	-1.500
pulp_2	0.098	0.0091	0.047
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pulp_4	85.000	29.0000	46.000

Inference for multiple regression

- ▶ Up to now, we have not assumed a full distribution for the errors (ϵ).
- ▶ To perform inference (calculate CI's and p-values), we make the additional assumption that the errors (ϵ) are normally distributed, implying that

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}).$$

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- ▶ Up to now, we have not assumed a full distribution for the errors (ϵ).
- ▶ To perform inference (calculate CI's and p-values), we make the additional assumption that the errors (ϵ) are normally distributed, implying that

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}).$$

- ▶ This implies that

$$\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}).$$

- ▶ As well as that

$$\frac{n-k}{\sigma^2} s^2 \sim \chi_{n-k}^2.$$

- ▶ We note again that $\hat{\beta}$ and s^2 are independent.

Confidence intervals for multiple regression

Using the previous results, a $100(1 - \alpha)\%$ confidence ellipsoid for β is given by

$$\frac{(\beta - \hat{\beta})' \mathbf{X}' \mathbf{X} (\beta - \hat{\beta})}{k s^2} \leq F_{k, n-k}^{1-\alpha},$$

implying that a confidence interval for β_j is

$$\hat{\beta}_j \pm t_{n-k}^{\alpha/2} \sqrt{s^2 (\mathbf{X}' \mathbf{X})_{jj}^{-1}}.$$

Hypothesis testing

Univariate

- ▶ To test the null hypothesis $H_0 : \beta_j = 0$, we use the test statistic

$$t = \frac{\hat{\beta}_j}{\sqrt{s^2(\mathbf{X}'\mathbf{X})_{jj}^{-1}}}.$$

- ▶ and reject at significance-level α if

$$|t| > t_{n-k}^{1-\alpha/2}.$$

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- ▶ and reject at significance-level α if

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Simultaneous

- ▶ To test the null hypothesis $H_0 : \beta^{(1)} = \mathbf{0}$ where $\beta^{(1)} : q \times 1$ and $\beta = (\beta^{(1)} \quad \beta^{(2)})'$, we use

$$F = \frac{\hat{\beta}^{(1)'} C_{11}^{-1} \hat{\beta}^{(1)}}{qs^2},$$

- ▶ where $(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{C} = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix}$.
- ▶ Accordingly, we reject at significance-level α if

$$F \geq F_{q,n-k}^{1-\alpha}.$$

Inference example

Consider the dataset of four wood-pulp variables and four paper properties:

```
data(  
  "data_tidy_paper",  
  package = "DataTidy23RodoHonsMult"  
)
```

We'll select only the first paper variable:

pulp_1	pulp_2	pulp_3	pulp_4	paper_1
-0.030	35.239	36.991	1.057	21.312
0.015	35.713	36.851	1.064	21.206
0.025	39.220	30.586	1.053	20.709
0.030	39.756	21.072	1.050	19.542

Inference example (cont.)

We extract the response matrix, $\mathbf{Y}$:

```
Y_mat <- data_tidy_paper[, "paper_1", drop = FALSE] |>  
  as.matrix()  
colnames(Y_mat) <- "paper_1"
```

—
x
—
21
21
21
20
—

Inference example (cont.)

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```
Y_mat <- data_tidy_paper[, "paper_1",  
  as.matrix()  
colnames(Y_mat) <- "paper_1"
```

```
—  
  x  
—  
21  
21  
21  
20  
—
```

We extract the data for the predictor matrix, $\mathbf{X}$, and prepend a column of 1s:

```
X_mat <- as.matrix(cbind(1,  
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    dplyr::select(starts_with("pulp"))  
))  
colnames(X_mat)[1] <- "intercept"
```

intercept	pulp_1	pulp_2	pulp_3	pulp_4
1	-0.030	35	37	1.1
1	0.015	36	37	1.1
1	0.025	39	31	1.1

Inference example (cont.)

Estimates

We first estimate the coefficients:

```
beta_vec <-  
  (solve(t(X_mat) %*% X_mat) %*%  
    t(X_mat)) %*% Y_mat  
colnames(beta_vec) <- "estimate"
```

	estimate
intercept	-74.000
pulp_1	-3.100
pulp_2	0.098
pulp_3	0.049
pulp_4	85.000

β_{pulp_1} confidence interval

To get the 95% confidence interval for the β_{pulp_1} coefficient, we first get the standard error:

```
XtX_mat <- t(X_mat) %*% X_mat  
XtX_mat_inv <- solve(XtX_mat)  
var_resp <- sum((Y_mat - X_mat %*% beta_vec)  
  (nrow(Y_mat) - ncol(X_mat)))  
sd_beta_pulp1 <- sqrt(var_resp) * sqrt(XtX_mat_inv[1,1])  
sd_beta_pulp1 |> signif(2)
```

```
[1] 1.9
```

Inference example (cont.)

β_{pulp_1} confidence interval (cont.)

We then get the relevant t -distributed quantile:

```
t_beta_pulp1 <- qt(0.975, nrow(Y_mat))  
t_beta_pulp1 |> signif(2)
```

```
[1] 2
```

The confidence interval is then:

```
(beta_vec[2] + c(-1, 1) * t_beta_pulp1  
  signif(2)
```

```
[1] -7.00  0.71
```

Hypothesis testing: $H_0 : \beta_{\text{pulp}_1} = 0$

The test statistic is given by

```
t_stat <- beta_vec[2] / sd_beta_pulp1  
t_stat |> signif(2)
```

```
[1] -1.6
```

The p-value is therefore:

```
(pt(-abs(t_stat),  
  nrow(Y_mat) - ncol(X_mat)) * 2  
  ) |> signif(2)
```

```
[1] 0.11
```

Inference example (cont.)

Hypothesis testing: $H_0 : \beta^{(1)} = \mathbf{0}$

We now test the hypothesis that

$$\beta^{(1)} = \beta_1 = \beta_3 = \beta_4 = 0.$$

We first extract $\mathbf{C}_1$:

```
C11_mat <- XtX_mat_inv[2:5, 2:5]
C11_mat <- C11_mat[-2, -2]
C11_mat |> signif(2)
```

	pulp_1	pulp_3	pulp_4
pulp_1	1.6000	0.00320	-1.300
pulp_3	0.0032	0.00015	0.049
pulp_4	-1.3000	0.04900	70.000

Hypothesis testing: $H_0 : \beta^{(1)} = \mathbf{0}$

We now extract the relevant estimates:

```
beta_vec_1 <- beta_vec[
  c(2, 4:5), , drop = FALSE
]
beta_vec_1 |> signif()
```

	estimate
pulp_1	-3.1203200
pulp_3	0.0494002
pulp_4	85.0761000

Inference example (cont.)

Hypothesis testing: $H_0 : \beta^{(1)} = 0$

The test statistic is then:

```
f_stat <- (t(beta_vec_1) %*%  
  solve(C11_mat) %*%  
  beta_vec_1) / length(beta_vec_1) /  
  var_resp  
f_stat |> signif(2)
```

```
          estimate  
estimate    16
```

Hypothesis testing: $H_0 : \beta^{(1)} = 0$

This yields the following p-value:

```
pf(f_stat, length(beta_vec_1),  
  nrow(Y_mat) - ncol(X_mat),  
  lower.tail = FALSE  
) |> signif(2)
```

```
          estimate  
estimate 1.6e-07
```


Inference for multivariate regression

- ▶ Multiple and multivariate regression produce *exactly* the same inference when coefficients are examined within the context of a single response variable.
- ▶ For example, suppose that we model BMI and blood pressure against age, weight and height:
 - ▶ The coefficients, p-values and confidence intervals for age, weight and height will be the same in both multiple (fitting two models) and multivariate regression.
 - ▶ The simultaneous hypothesis tests will be the same when the considering only groups of coefficients with respect to a particular response variable. As an example, the p-value for the hypothesis that neither age nor weight has an effect on BMI will be the same in both models.
- ▶ The main difference of interest is that we can perform simultaneous hypothesis tests for regression coefficients across different response variables in multivariate regression.
 - ▶ To continue the example above, in multivariate regression we can test that the effects of age on BMI and blood pressure are both zero.

Hypothesis testing in multivariate regression

Suppose we wish to test that

$$\mathcal{B}^{(1)} = \mathbf{0},$$

where

$$\mathcal{B} = \begin{pmatrix} \mathcal{B}^{(1)} \\ \mathcal{B}^{(2)} \end{pmatrix}$$

and $\mathcal{B}^{(1)} : q \times r$.

This is the hypothesis that the first q predictors have no effect on any of the r responses.

Hypothesis testing in multivariate regression

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$$\mathcal{B}^{(1)} = \mathbf{0},$$

where

$$\mathcal{B} = \begin{pmatrix} \mathcal{B}^{(1)} \\ \mathcal{B}^{(2)} \end{pmatrix}$$

and $\mathcal{B}^{(1)} : q \times r$.

This is the hypothesis that the first q predictors have no effect on any of the r responses.

- ▶ We'll need to fit a model without the first q predictors and compare the fit of the two models.
- ▶ Let's define $\mathbf{X}^{(2)} : n \times (k - q)$ for $i = 1, 2$ as follows:

$$\mathbf{X} = (\mathbf{X}^{(1)} \quad \mathbf{X}^{(2)})$$

Hypothesis testing in multivariate regression

- ▶ Under the implicit assumption that $\mathcal{B}^{(1)} = \mathbf{0}$, we obtain the estimated regression parameters as

$$\hat{\mathcal{B}}^{(2)} = (\mathbf{X}^{(2)'} \mathbf{X}^{(2)})^{-1} \mathbf{X}^{(2)'} y,$$

- ▶ and the MLE variance-covariance matrices as

$$\hat{\Sigma}_{\text{MLE}} = \frac{1}{n} (y - \mathbf{X} \hat{\mathcal{B}})' (y - \mathbf{X} \hat{\mathcal{B}}).$$

- ▶ and

$$\hat{\Sigma}_{\text{MLE}}^{(2)} = \frac{1}{n} (y - \mathbf{X}^{(2)} \hat{\mathcal{B}}^{(2)})' (y - \mathbf{X}^{(2)} \hat{\mathcal{B}}^{(2)}).$$

Hypothesis testing in multivariate regression

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- ▶ and

$$\hat{\Sigma}_{\text{MLE}}^{(2)} = \frac{1}{n} (\mathbf{y} - \mathbf{X}^{(2)} \hat{\mathcal{B}}^{(2)})' (\mathbf{y} - \mathbf{X}^{(2)} \hat{\mathcal{B}}^{(2)}).$$

- ▶ The test statistic is then

$$\begin{aligned} \Lambda &= \frac{\max_{\mathcal{B}^{(2)}, \Sigma^{(2)}} L(\mathcal{B}^{(2)}, \Sigma^{(2)})}{\max_{\mathcal{B}, \Sigma} L(\mathcal{B}, \Sigma)}, \\ &= \left(\frac{|\hat{\Sigma}|}{|\hat{\Sigma}^{(2)}|} \right)^{n/2}. \end{aligned}$$

- ▶ For $n - r$ and $n - k$ sufficiently large, the following approximation holds:

$$-[n - k - 0.5(r - p + q + 1)] \ln(\Lambda^{2/n}) \sim \chi_{rq}^2.$$

Multivariate inference example

Using the paper dataset, we will test whether pulp_2 or pulp_3 have an effect on any paper type.

Loading the data as follows:

```
data(  
  "data_tidy_paper",  
  package = "DataTidy23RodoHonsMult"  
)
```

pulp_1	pulp_2	pulp_3	pulp_4	paper_1	paper_2	paper_3	paper_4
-0.030	35.239	36.991	1.057	21.312	7.039	5.326	0.932
0.015	35.713	36.851	1.064	21.206	6.979	5.237	0.871
0.025	39.220	30.586	1.053	20.709	6.779	5.060	0.742
0.030	39.756	21.072	1.050	19.542	6.601	4.479	0.513

Multivariate example (cont.)

We extract the response matrix, $\mathbf{Y}$:

```
Y_mat <- data_tidy_paper |>  
  dplyr::select(-starts_with("pulp")) |>  
  as.matrix()
```

paper_1	paper_2	paper_3	paper_4
21	7.0	5.3	0.93
21	7.0	5.2	0.87
21	6.8	5.1	0.74
20	6.6	4.5	0.51

Multivariate example (cont.)

We extract the response matrix, $\mathbf{Y}$:

```
Y_mat <- data_tidy_paper |>
  dplyr::select(-starts_with("pulp"))
  as.matrix()
```

paper_1	paper_2	paper_3	paper_4
21	7.0	5.3	0.93
21	7.0	5.2	0.87
21	6.8	5.1	0.74
20	6.6	4.5	0.51

We extract the data for the predictor matrix, $\mathbf{X}$, and prepend a column of 1s:

```
X_mat <- as.matrix(cbind(1,
  data_tidy_paper |>
    dplyr::select(starts_with("pulp")))
))
colnames(X_mat)[1] <- "intercept"
```

intercept	pulp_1	pulp_2	pulp_3	pulp_4
1	-0.030	35	37	1.1
1	0.015	36	37	1.1
1	0.025	39	31	1.1

Multivariate example (cont.)

We extract the data for the predictor matrix excluding pulp_2 and pulp_3, $\mathbf{X}^{(2)}$:

```
X_mat_2 <- X_mat[  
  , -(3:4), drop = FALSE  
]
```

intercept	pulp_1	pulp_4
1	-0.030	1.1
1	0.015	1.1
1	0.025	1.1

We obtain estimated regression coefficients under both the full and reduced models:

```
B_mat <- (  
  solve(t(X_mat) %*% X_mat) %*%  
  t(X_mat)) %*% Y_mat
```

```
B_mat_2 <- (  
  solve(t(X_mat_2) %*% X_mat_2) %*%  
  t(X_mat_2)) %*% Y_mat
```

Multivariate example (cont.)

We now calculate the sample variance-covariance matrices under the full and reduced models:

```
Sigma_mat <-  
  t((Y_mat - X_mat %*% B_mat)) %*%  
    (Y_mat - X_mat %*% B_mat) /  
    (nrow(Y_mat))
```

```
Sigma_mat_2 <-  
  t((Y_mat - X_mat_2 %*% B_mat_2)) %*%  
    (Y_mat - X_mat_2 %*% B_mat_2) /  
    (nrow(Y_mat))
```

► This yields the following test statistic:

```
n <- nrow(Y_mat)  
r <- ncol(Y_mat)  
q <- ncol(X_mat) - ncol(X_mat_2)  
k <- ncol(X_mat)  
p <- k - 1  
test_stat <-  
  -(n - k - 0.5 * (r - p + q + 1)) *  
  log(  
    det(Sigma_mat) / det(Sigma_mat_2)  
  )  
test_stat |> signif(2)
```

[1] 40

with p-value

```
pchisq(test_stat, r * (p - q), lower.tail = FALSE)  
|> signif(2)
```

Principal components regression (PCR)

- ▶ In PCR, we replace the original p (non-intercept) predictors with $m \leq p$ principal components.
- ▶ Motivations:
 - ▶ Avoid multicollinearity
 - ▶ Reduce overfitting
 - ▶ Use fewer predictors
 - ▶ Improve interpretability

Very straightforward...

1. Calculate the principal components of the (non-intercept) predictors.
2. Fit a multiple regression model using the first m principal components as predictors.

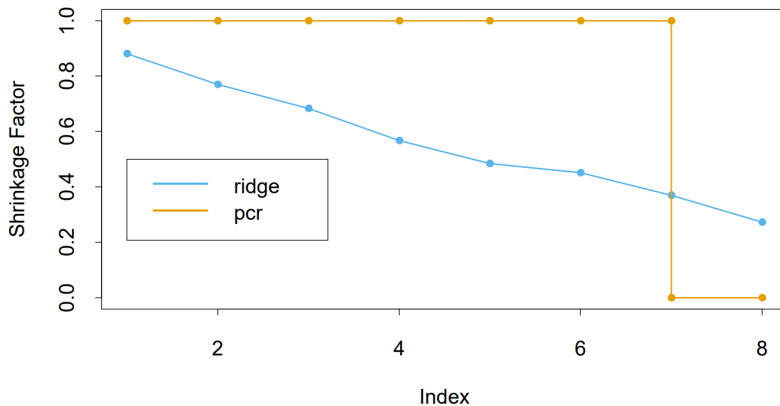
Very straightforward...

1. Calculate the principal components of the (non-intercept) predictors.
2. Fit a multiple regression model using the first m principal components as predictors.

If the number of principal components is pre-specified, then all the inferential methods of multiple regression apply.

Relationship with ridge regression

- PCR is akin to a rough version of ridge regression (Hastie, Tibshirani, and Friedman 2009):



Selecting the number of principal components

- ▶ One can use the same reasoning as before (scree plots, proportion of variation captured), or cross-validation (or R^2 , but this isn't really done).
- ▶ Note that when performing cross validation, the loadings must be computed afresh within each fold.

Partial least squares (PLS)

- ▶ Partial least squares (PLS) is a method that is similar to PCR, in that it takes linear combinations of the input variables.
- ▶ However, instead of choosing the linear combinations to maximise the variance, it chooses them to maximise (something like) the covariance of the resultant variables with the response.

Partial least squares algorithm (Hastie, Tibshirani, and Friedman 2009)

-
1. Standardize each $\mathbf{x}_j$ to have mean zero and variance one. Set $\hat{\mathbf{y}}^{(0)} = \bar{y}\mathbf{1}$, and $\mathbf{x}_j^{(0)} = \mathbf{x}_j$, $j = 1, \dots, p$.
 2. For $m = 1, 2, \dots, p$
 - (a) $\mathbf{z}_m = \sum_{j=1}^p \hat{\varphi}_{mj} \mathbf{x}_j^{(m-1)}$, where $\hat{\varphi}_{mj} = \langle \mathbf{x}_j^{(m-1)}, \mathbf{y} \rangle$.
 - (b) $\hat{\theta}_m = \langle \mathbf{z}_m, \mathbf{y} \rangle / \langle \mathbf{z}_m, \mathbf{z}_m \rangle$.
 - (c) $\hat{\mathbf{y}}^{(m)} = \hat{\mathbf{y}}^{(m-1)} + \hat{\theta}_m \mathbf{z}_m$.
 - (d) Orthogonalize each $\mathbf{x}_j^{(m-1)}$ with respect to $\mathbf{z}_m$: $\mathbf{x}_j^{(m)} = \mathbf{x}_j^{(m-1)} - [\langle \mathbf{z}_m, \mathbf{x}_j^{(m-1)} \rangle / \langle \mathbf{z}_m, \mathbf{z}_m \rangle] \mathbf{z}_m$, $j = 1, 2, \dots, p$.
 3. Output the sequence of fitted vectors $\{\hat{\mathbf{y}}^{(m)}\}_1^p$. Since the $\{\mathbf{z}_\ell\}_1^m$ are linear in the original $\mathbf{x}_j$, so is $\hat{\mathbf{y}}^{(m)} = \mathbf{X} \hat{\beta}^{\text{pls}}(m)$. These linear coefficients can be recovered from the sequence of PLS transformations.
-

PLS versus PCR (Hastie, Tibshirani, and Friedman 2009)

- ▶ PCR and PLS optimise different objective functions:

the m th principal component direction v_m solves:

$$\begin{aligned} & \max_{\alpha} \text{Var}(\mathbf{X}\alpha) \\ & \text{subject to } \|\alpha\| = 1, \alpha^T \mathbf{S} v_{\ell} = 0, \ell = 1, \dots, m-1, \end{aligned} \tag{3.63}$$

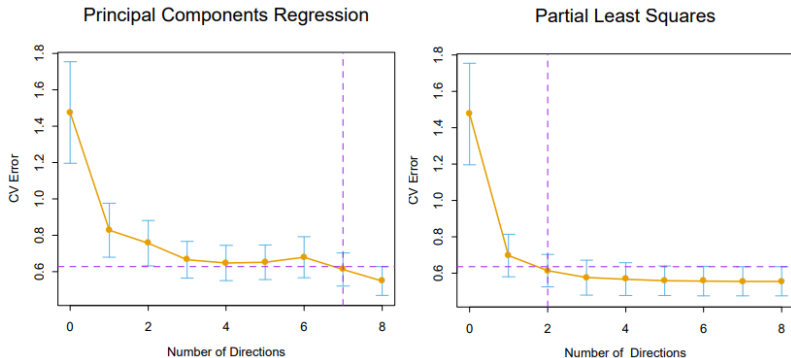
where $\mathbf{S}$ is the sample covariance matrix of the $\mathbf{x}_j$.

The m th PLS direction $\hat{\varphi}_m$ solves:

$$\begin{aligned} & \max_{\alpha} \text{Corr}^2(\mathbf{y}, \mathbf{X}\alpha) \text{Var}(\mathbf{X}\alpha) \\ & \text{subject to } \|\alpha\| = 1, \alpha^T \mathbf{S} \hat{\varphi}_{\ell} = 0, \ell = 1, \dots, m-1. \end{aligned} \tag{3.64}$$

Selecting the number of components in PCR and PLS (Hastie, Tibshirani, and Friedman 2009)

- ▶ The number of components can be chosen by cross-validation.
- ▶ Hastie, Tibshirani, and Friedman (2009) chose the number of components within one standard error of the minimum cross-validated error:



Resources

- ▶ Source code is on [GitHub](#).
 - ▶ Suppressed code (e.g. to create figures) is available there.

References

- Hastie, Trevor., Robert. Tibshirani, and J. H. (Jerome H.) Friedman. 2009. *The Elements of Statistical Learning : Data Mining, Inference, and Prediction*. 2nd ed. Springer Series in Statistics. New York: Springer.
- Johnson, Richard A., and Dean W. Wichern. 2007. *Applied Multivariate Statistical Analysis*. 6th ed. Upper Saddle River, NJ: Pearson Education Inc.